CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2013 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Define Database and Database Management System.

Definition: A Database is an organized collection of logically related data. A DBMS is the software interface that controls data storage, retrieval, and access permissions.

Q2 (3M) What is Data Redundancy?

Definition: Unnecessary duplication of identical data items across multiple database files, causing wasted storage and risk of inconsistency.

Q3 (3M) Define Primary Key.

Definition: A unique attribute chosen to identify tuples in a relation. Cannot accept duplicate values or NULL.

Q4 (3M) What is Relational Data Model?

Definition: A data model representing data in terms of mathematical relations (2D tables) composed of rows (tuples) and columns (attributes).

Q5 (3M) What is a Transaction?

Definition: A logical unit of database processing that includes one or more database access operations, executed as an indivisible atomic action.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain the Three-Level ANSI/SPARC Database Architecture.

Architecture: External, Conceptual, and Internal levels providing logical and physical data independence.

Q7 (7.5M) Explain Hashing and all collision resolution techniques in detail.

Collision Resolution: Detailed explanation of Linear Probing, Quadratic Probing, Chaining with overflow buckets, and Extendible Hashing.

Q8 (7.5M) What do you understand by Transaction System? Explain ACID properties.

Transaction Management: Atomicity, Consistency, Isolation, and Durability guaranteed through concurrency control and recovery.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) What is Normalization? Explain in detail 1st, 2nd, and 3rd Normal Form with importance.

Complete Normalization: 1NF (atomic values), 2NF (elimination of partial dependencies), and 3NF (elimination of transitive dependencies) with complete table decomposition walkthroughs.

Q10 (15M) **Explain Relational Algebra Operations with expressions and examples.**

Relational Algebra: Complete coverage of Selection, Projection, Cartesian Product, Union, Difference, and Natural Join.

Q11 (15M) **Explain ER Diagram concepts: Entity Sets, Weak Entities, Attributes, Relationships, and Cardinality. Draw ER diagram for Banking System.**

ER Model: Detailed conceptual design for Customer, Account, Branch, and Loan entities with conversion to relational tables.

Q12 (15M) **Explain File Organization Techniques: ISAM, B-Tree, and B+ Tree with diagrams.**

File Organization: Detailed structural differences and access speeds between Indexed Sequential, B-Tree, and B+ Tree structures.

Q13 (15M) **Explain Concurrency Control Protocols: Two-Phase Locking (2PL) and Timestamp Ordering.**

Concurrency Control: Strict 2PL, Rigorous 2PL, Thomas Write Rule, and timestamp-based conflict resolution.
